

Experiment-01

NMOS Characterization

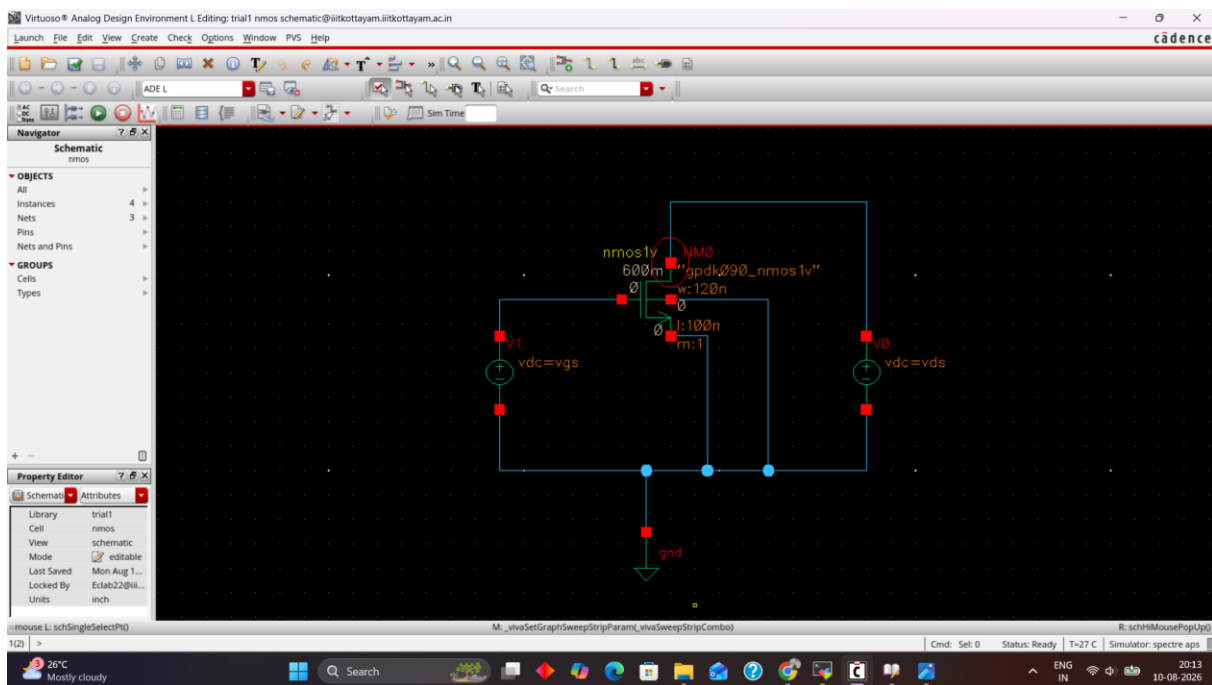
Analog Electronics Lab

Name: ALLWIN STANLEY

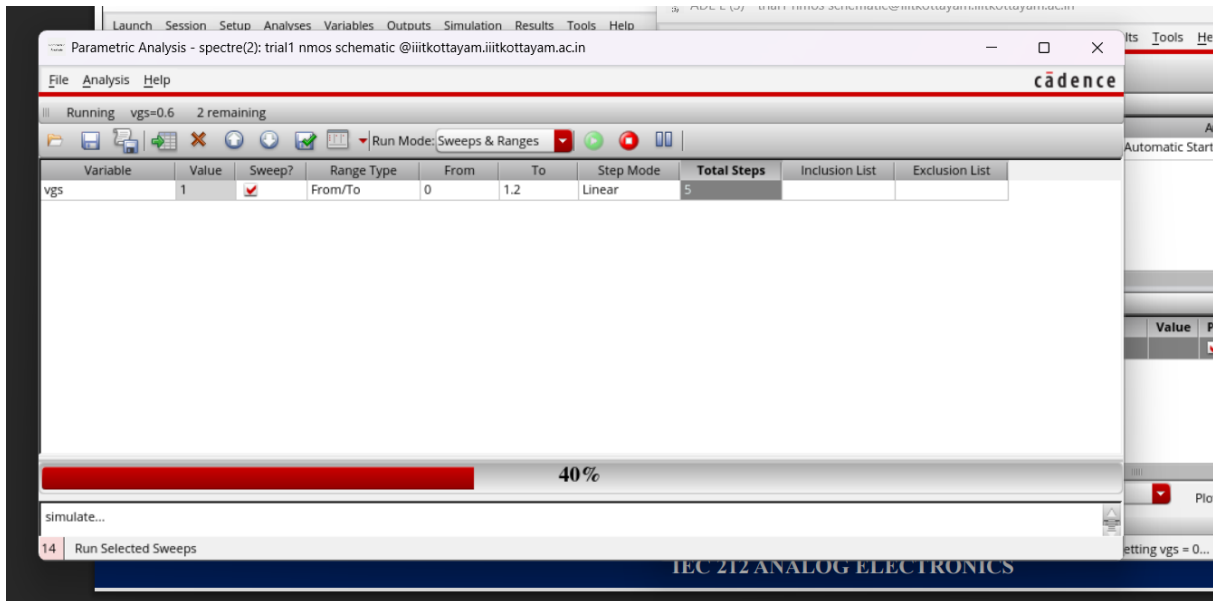
Roll No.: 2025BEC0067

GitHub: https://github.com/allwinec/AnalogElectronicsLab_2025BEC0067

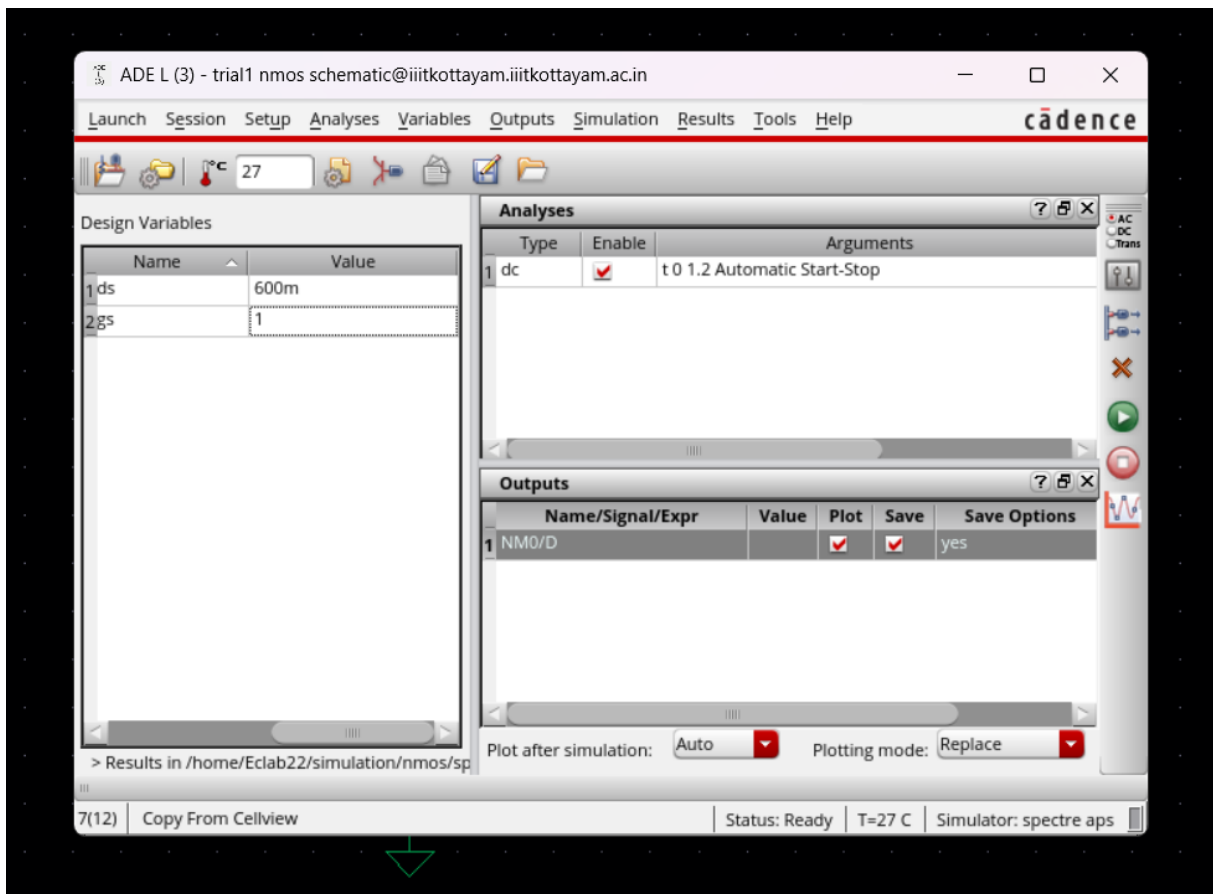
Schematic



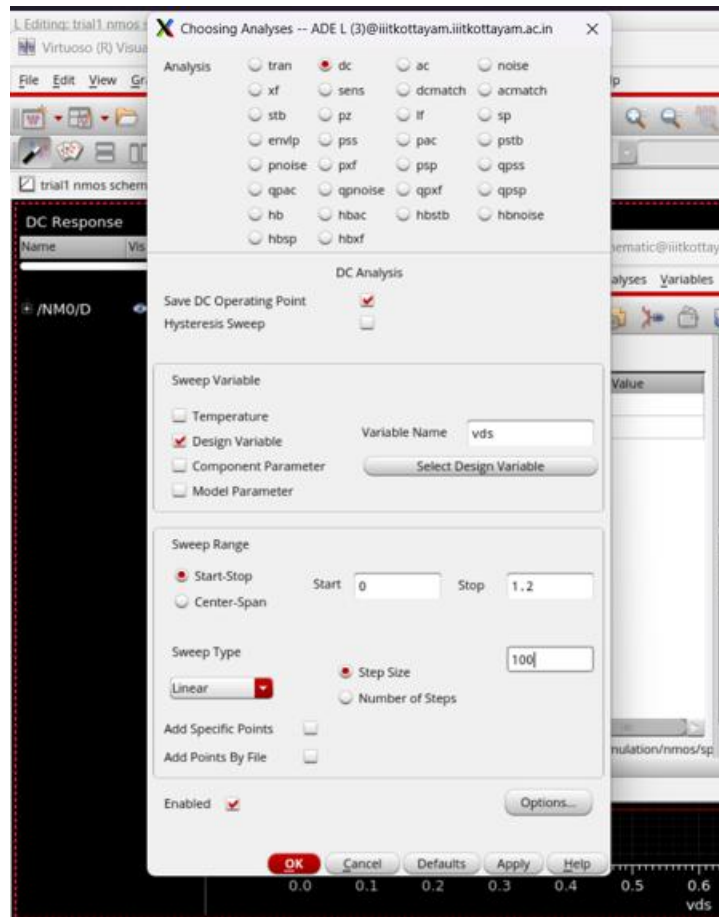
Simulation Setup



Parametric Sweep



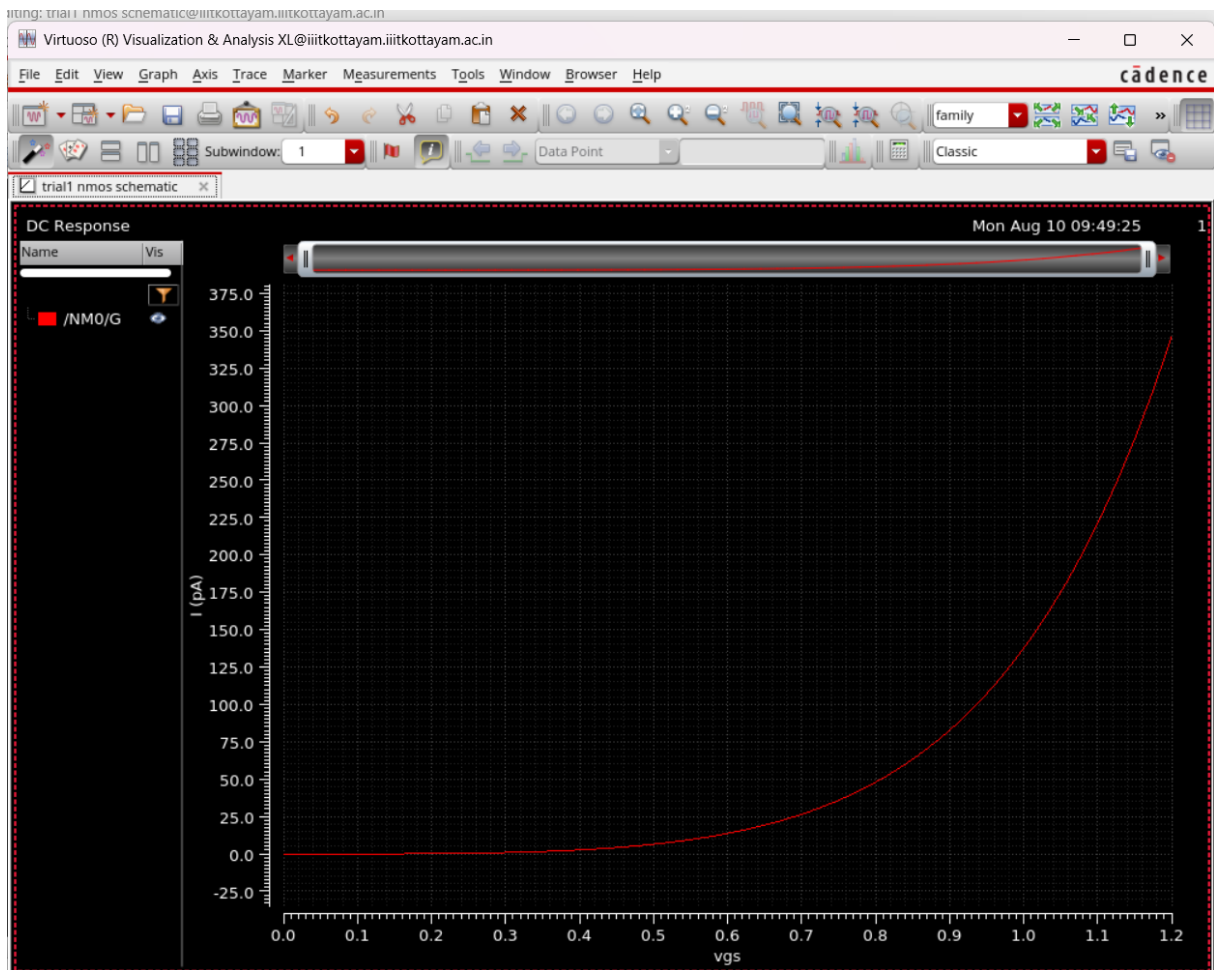
DC Operating Point



DC Operating Point Result

Results Display Window@iiitkottayam.iiitkottayam.ac.in	
Window Expressions Info Help	cadence
signal	OP("/NM0" "?")
qgi	1.17438f
qinv	2.85359m
qsi	-682.68a
qsrco	-1.04711f
region	2
reversed	0
ron	1.027K
rout	3.4383K
sdint	2.14748G
sdop	-1
self_gain	3.12136
tau1	NaN
tk	NaN
trise	NaN
ueff	16.7749m
vbs	0
vdb	600m
vds	600m
vdsat	402.099m
vdsat_marg	NaN
vdss	402.099m
vearly	2.00873
vfbeff	-899.062m
vgb	1
vgd	400m
vgs	1
vgsteff	770.042m
vgt	788.53m
vsat_marg	197.901m
vsb	-0
vth	211.47m
vth0	NaN
vth_drive	NaN

VGS Transfer Characteristic



VDS Output Characteristics

